	LINEAR PROGRAMMING PROBLEM - CONSTRAINT OPTIMIZATION

AIM:

To solve a Linear Programming Problem (LPP) for constraint optimization using Python.

PROCEDURE:

1. **Install Required Libraries:**
   - Make sure you have numpy and scipy installed. You can install them using pip if they are not already installed.

l **bash**

l **Copy**

l **pip install numpy scipy**

2. **Define the Problem:**
   - Identify the objective function to maximize or minimize.
   - Determine the constraints (equality and inequality).
3. **Set Up the Problem in Python:**
   - Use the scipy.optimize.linprog function to set up and solve the LPP.
4. **Run the Code:**
   - Execute the Python code to get the optimized result.
5. **Analyze the Output:**
   - Interpret the results and validate the solution against the problem constraints.

PROBLEM-1:

Maximize

$Z = 3x + 2y$

Subject to:

$x + y \leq 4$

$x \leq 2$
$y \leq 3$
$x, y \geq 0$

Code:

```
import pulp
lp = pulp.LpProblem("LPP1", pulp.LpMaximize) x

= pulp.LpVariable('x', lowBound=0)

y = pulp.LpVariable('y', lowBound=0)

lp += 3*x +
2*y lp += x
+ y <= 4 lp
+= x <= 2
lp += y <=

3 lp.solve()

print(x.value(), y.value(), pulp.value(lp.objective))
```

OUTPUT:

Optimal
Value (Z) :
10.0 x = 2.0
y = 2.0

RESULT:

Thus, the Linear Programming Problems for constraint optimization using Python were successfully solved, and the solutions were obtained for all the practice problems provided.